🛠️ Part 1: Technology Stack Summary

Frontend (The User Interface)

- Framework: Next.js 15+ (React) – Chosen for high performance, SEO friendliness, and fast page transitions.
- Styling: Tailwind CSS – For the premium, custom "Glassmorphism" design you see in the Admin Portal.
- Icons: Lucide React – A modern, consistent icon set for the dashboard and chat buttons.
- State Management: React Hooks (useState/useEffect) for real-time UI updates.

Backend (The Engine)

- Language: Python 3.12 – The industry standard for AI and data processing.
- API Framework: FastAPI – An extremely fast, high-performance web framework for handling chat requests and admin actions.
- Database ORM: SQLAlchemy – Handles all data communication (users, messages, documents) securely.

AI & Intelligence (The Brain)

- Large Language Model (LLM): OpenAI GPT-4o – Powers the natural language conversations.
- RAG (Retrieval Augmented Generation): LangChain – The logic that allows the AI to "read" your custom PDF documents before answering.
- Vector Processing: FAISS – A high-speed mathematical library that searches through thousands of PDF pages in milliseconds.

Persistence (The Content)

- Database: MySQL – Stores user profiles, chat history, and document metadata.
- File Storage: Local filesystem for PDF uploads.

🚀 Part 2: Hosting Strategy (For your Client)

To host this for a client so it's accessible 24/7 on the web, I recommend the following "Gold Standard" setup:

1. The Frontend (Vercel)

- Why: Created by the makers of Next.js. It's the fastest and easiest way to host your UI.
- Cost: Free tier available; Pro is ~$20/mo.
- Website: vercel.com

2. The Backend (Railway or Render)

- Why: These platforms are perfect for Python/FastAPI. They handle "scaling" automatically if the client gets a lot of traffic.

- Setup: You simply link your GitHub repository, and they deploy it automatically.
- Cost: ~$5–$10/mo depending on usage.
- Website: railway.app or render.com

3. The Database (Managed MySQL)

- Why: You shouldn't host the database on your own laptop for a client. Use a managed service like Railway MySQL or PlanetScale.
- Cost: Usually included in the backend hosting price or a small extra fee.

4. The Domain (e.g., support.clientname.com)

- Why: To make it look professional.
- Service: Namecheap, Google Domains, or GoDaddy.

📋 Moving to Production (Checklist)

1. Environment Variables: Move all keys (OpenAI, DB URL) into the hosting provider's "Environment Variables" section.
2. SSL/HTTPS: Vercel and Railway provide this automatically (it adds the "locked" icon in the browser).
3. API Key Management: Ensure the client has their own OpenAI account and API key for billing.